Uniruled Fourfolds and Mukai Lattice Cycle Reductions in Lean 4: Conditional Cycle-Class Frameworks

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/uniruled-fourfolds
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Abstract

Standalone Lean 4 formalization (‘UniruledFourfolds.lean‘, ‘UniruledFourfolds/Blueprint.lean‘, ‘UniruledFourfolds/MukaiLattice.lean‘) and Python Mukai lattice signature/discriminant calculator (‘scripts/mukai_lattice_calc.py‘) for conditional cycle-class reductions on smooth projective uniruled fourfolds. Formalizes how middle Hodge classes reduce under cylinder/correspondence maps when a uniruled parametrization hypothesis holds.

1 Scope, Scientific Honesty, and Non-Claim Boundary

All cycle-class surjectivity deductions in this repository are conditional on explicit geometric correspondence hypotheses; this repository does not claim an unconditional proof of any open Hodge surjectivity conjecture.

- **Repository:** `https://github.com/BenFrohman/uniruled-fourfolds`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652974`
- **Earliest Git Commit SHA:** `18b62b11b26338ecab62426eaf4844cc57e8353b` (2026-10-01 04:51:29 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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The Clay Mathematics Institute Hodge conjecture is open. This repository does not resolve it.

What the literature already contains, and what this repository does not formalize, is recorded in docs/special-cases.md. Short form:

Statement	Status
Hodge conjecture for a general smooth complex projective variety	**Open.**

| Rational Hodge conjecture for fourfolds covered by rational curves | **Theorem** of Conte–Murre, Math. Ann. **238** (1978). Not proved in this repository. |

| Smooth cubic fourfolds | **Theorem** of Zucker, Compositio Math. **34** (1977). They are also Fano, so Conte–Murre applies. |

| Smooth complex Fano fourfolds | **Corollary** of Conte–Murre plus Kollár–Miyaoka–Mori, J. Diff. Geom. **36** (1992). Not a case beyond uniruled fourfolds. |

| Rationality of a general cubic fourfold, Kuznetsov's conjecture | **Open.** |

The uniruled reduction uses a rational curve through a general point. It stops there. It is not a route to the general fourfold.

'Blueprint.lean' is a conditional sketch: *if* a Bloch–Srinivas package is supplied for a uniruled fourfold, the file *states* that codimension-2 cycle-class surjectivity would reduce to the classical Lefschetz theorem on (1,1)-classes on a threefold. The proof is 'sorry'. Elaborating the file only checks that the statement is well-formed.

'MukaiLattice.lean' defines the integer form

\{}[

\{}langle v, v \{}rangle = v_2^2 - 2 v_0 v_4, \{}qquad \{}text{primitive} \{}iff \{}gcd(|v_0|, |v_2|, |v_4|) = 1

\{}]

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/uniruled-fourfolds` is audited via `titan-audit` (`audit/TITAN_AUDIT_C`
and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`)
signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
.github/ISSUE_TEMPLATE/peer_review.md	782 B	5f631a14dc34eac3...
.github/workflows/verify.yml	1183 B	352bc80124ce57e0...
.gitignore	205 B	9b49bb8918dfc2a6...
COPYRIGHT	273 B	6090f1609bc4aa06...
GEOMETRY.md	15556 B	21a7d3f8dd690bbb...
LICENSE	11355 B	9da72c2bee4f20ba...
NOTICE	485 B	db024313d28bb067...
README.md	4954 B	810976598544dc50...
RELEASES.md	1294 B	ac033062c1d4633a...
RELEASE_NOTES.md	1984 B	272a88719103d22b...
UniruledFourfolds/Blueprint.lean	2831 B	32254baf297f4cf5...
UniruledFourfolds/MukaiLattice.lean	26313 B	631233c45cf3ab94...
deploy_potluck.sh	1367 B	ae18cf897a90c640...
docs/special-cases.md	7354 B	61fcbbca0a53402b...
lake-manifest.json	154 B	f74dbc7f897503a6...
lakefile.lean	666 B	e088462a2c63308f...
lean-toolchain	25 B	d5edba4e4b8faad9...
plots/render_formulas.py	16937 B	b358af22bf95d7b7...
potluck/finding_mod_gaussian_shannon_entropy_001.py	1518 B	9a2b1f88cd78a83c...
potluck/finding_mod_grand_finale_999.py	1869 B	71a6d19b18ad02b4...
potluck/finding_mod_ksat_shift_014.py	1341 B	87c7572da9db5958...
potluck/finding_mod_navier_stokes_007.py	2881 B	8c5a16f66840156d...
potluck/finding_mod_navier_stokes_pressure_010.py	3689 B	f002959ece6fcfd4...
potluck/finding_mod_nmf_factor_003.py	2538 B	9a2e3f49772cd2bb...
potluck/finding_mod_sat_contour_016.py	991 B	2288e0f6fdecde34...
potluck/finding_mod_sat_entropy_020.py	1376 B	f3c482adc82e206c...
potluck/finding_mod_sat_phase_transition_012.py	2208 B	c58c3de5e59681ac...
potluck/finding_mod_sat_wireframe_018.py	728 B	8393c9d79e0b9c07...
potluck/finding_mod_spectral_map_002.py	2929 B	9d2a7ab6b0a0a49e...
potluck/sync_manifest.py	3454 B	449675f78065697d...
scripts/pre-commit	1076 B	e9b21c0ad9afb183...
scripts/pre-push	540 B	3cdbc6e44a570d83...
validate_mukai.py	6552 B	de91900884db478b...
verify_lean.sh	890 B	e191c4ef992a1738...

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